VALUATIVE CRITERION VARIANT FOR UNIVERSAL CLOSEDNESS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3088-3113 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 3115-3163, the complete author proof (47 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 22 of queue_stacks-stacks-more-morphisms.json (pb=2233, stmt 3088-3113, proof 3115-3163).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned. Named omission: inheritance of the fill-in property by products and by the reduced replacement - same-flavor mechanical verifications done explicitly for the base-change case (artin #8 precedent).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma lemma-reach-point-closure-Noetherian: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2800–2813. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0H2A: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLA: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CLE: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CLD: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 3163–3163. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume that $\mathcal{Y}$ is locally Noetherian and that f is of finite type. If given any 2-commutative diagram*

$$\begin{array}{ccc} \mathrm{Spec}(K) & \xrightarrow{x} & \mathcal{X} \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & \xrightarrow{y} & \mathcal{Y} \end{array}$$

where A is a discrete valuation ring with field of fractions K and $\gamma : y \circ j \to f \circ x$ there exist an extension K'/K of fields, a valuation ring $A' \subset K'$ dominating A such that the category of dotted arrows for the induced diagram

$$\begin{array}{ccc} \mathrm{Spec}(K') & \xrightarrow{} & \mathcal{X} \\ {\scriptstyle j'}\downarrow & \overset{x'}{\nearrow} & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A') & \xrightarrow{y'} & \mathcal{Y} \end{array}$$

with induced 2-arrow $\gamma' : y' \circ j' \to f \circ x'$ is nonempty (Morphisms of Stacks, Definition 0CLA), then f is universally closed.

Proof. Let $V \to \mathcal{Y}$ be a smooth morphism where V is an affine scheme. The category of dotted arrows behaves well with respect to base change (Morphisms of Stacks, Lemma 0CLE). Hence the assumption on existence of dotted arrows (after extension) is inherited by the morphism $\mathcal{X} \times_{\mathcal{Y}} V \to V$. Therefore the assumptions of the lemma are satisfied for the morphism $\mathcal{X} \times_{\mathcal{Y}} V \to V$. Hence we may assume $\mathcal{Y}$ is an affine scheme.

Assume $\mathcal{Y} = Y$ is a Noetherian affine scheme. (From now on we no longer have to worry about the 2-arrows γ and γ', see Morphisms of Stacks, Lemma 0CLD.) To prove that f is universally closed it suffices to show that $|\mathcal{X} \times \mathbf{A}^n| \to |Y \times \mathbf{A}^n|$ is closed for all n by Limits of Stacks, Lemma 0H2A. Since the assumption in the lemma is inherited by the product morphism $\mathcal{X} \times \mathbf{A}^n \to Y \times \mathbf{A}^n$ (details omitted) we reduce to proving that $|\mathcal{X}| \to |Y|$ is closed.

Assume Y is a Noetherian affine scheme. Let $T \subset |\mathcal{X}|$ be a closed subset. We have to show that the image of T in $|Y|$ is closed. We may replace $\mathcal{X}$ by the reduced induced closed subspace structure on T; we omit the verification that property on the existence of dotted arrows is preserved by this replacement. Thus we reduce to proving that the image of $|\mathcal{X}| \to |Y|$ is closed.

Let $y \in |Y|$ be a point in the closure of the image of $|\mathcal{X}| \to |Y|$. By Lemma lemma-reach-point-closure-Noetherian we may choose a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow{\scriptstyle f} \\ \mathrm{Spec}(A) & \longrightarrow & Y \end{array}$$

where A is a discrete valuation ring and K is its field of fractions mapping the closed point of $\mathrm{Spec}(A)$ to y. It follows immediately from the assumption in the lemma that y is in the image of $|\mathcal{X}| \to |Y|$ and the proof is complete. $\square$